

GEOMETRY

Tuesday, June 23, 2026 — 1:15 to 4:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 35 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice ...

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

Tear Here

Tear Here

Reference Sheet for Geometry

Volume	Cylinder	$V = Bh$ where B is the area of the base
	General Prism	$V = Bh$ where B is the area of the base
	Sphere	$V = \frac{4}{3}\pi r^3$
	Cone	$V = \frac{1}{3}Bh$ where B is the area of the base
	Pyramid	$V = \frac{1}{3}Bh$ where B is the area of the base

Part I

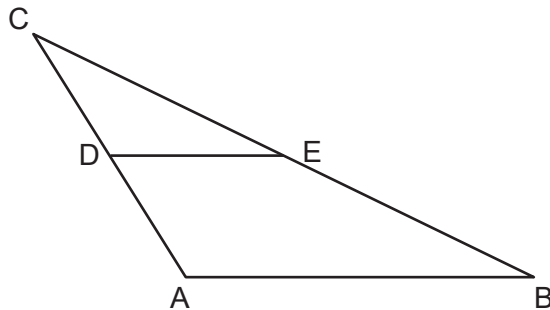
Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 Which transformation would result in the area of a rectangle's image being different from the area of its pre-image?

- (1) a reflection over the y -axis
- (2) a translation 4 units to the right
- (3) a rotation of 90° counterclockwise about the origin
- (4) a vertical stretch of scale factor 3 with respect to $y = 0$

- 2 In the diagram below of $\triangle ABC$, points D and E are the midpoints of $\overline{CA}$ and $\overline{CB}$, respectively.

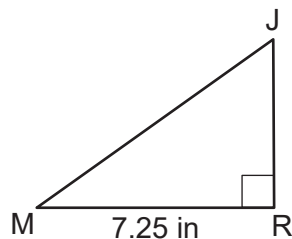


Which statement must always be true?

- (1) $DE = \frac{1}{2}AB$
- (2) $DE = \frac{1}{2}AC$
- (3) $AD = \frac{1}{2}AB$
- (4) $AB = \frac{1}{2}DE$

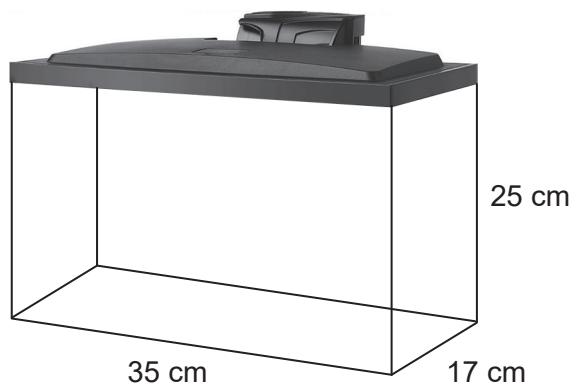
- 3 In triangle RJM below, $m\angle R = 90^\circ$ and $MR = 7.25$ inches.

Use this space for
computations.



If the measure of angle M is 35° , what is the length of $\overline{MJ}$, to the nearest hundredth of an inch?

- (1) 4.16 (3) 8.85
(2) 5.94 (4) 12.64
- 4 A fish tank in the shape of a rectangular prism with a length of 35 cm, width of 17 cm, and a height of 25 cm is shown below.



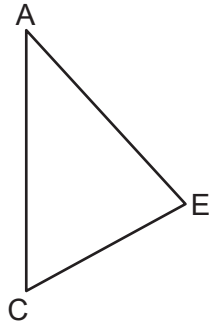
If the fish tank is filled with water to a height 3 centimeters from the top, how many liters of water are in this tank, to the nearest liter?

[1 liter = 1000 cubic centimeters]

- (1) 10 (3) 15
(2) 13 (4) 17

Use this space for
computations.

- 5 Triangle ACE is drawn below. Triangle ACE is mapped onto triangle AXE after a reflection over side $\overline{AE}$. Triangle AXE is then mapped onto triangle LXE after a reflection over side $\overline{XE}$.

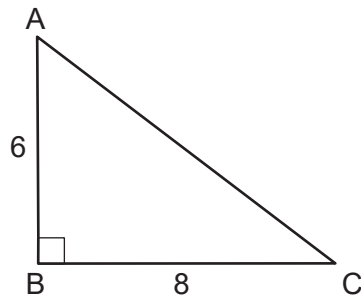


Which side of $\triangle LXE$ is the image of $\overline{AC}$?

- | | |
|---------------------|---------------------|
| (1) $\overline{LE}$ | (3) $\overline{XE}$ |
| (2) $\overline{AX}$ | (4) $\overline{LX}$ |
- 6 The lines whose equations are represented by $y = -\frac{1}{2}x + 2$ and $x + 2y = 8$ are
- (1) parallel
 - (2) perpendicular
 - (3) the same line
 - (4) neither parallel nor perpendicular

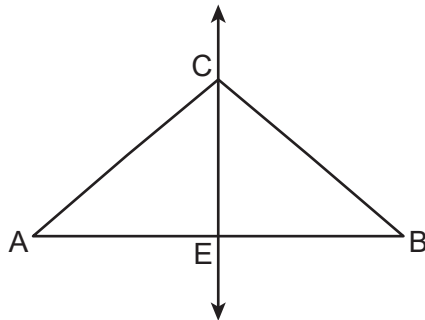
Use this space for
computations.

- 7 Right triangle ABC below has legs whose lengths are 6 and 8.



What is the volume of the three-dimensional object formed by continuously rotating $\triangle ABC$ about $\overline{AB}$?

- (1) 96π (3) 288π
(2) 128π (4) 384π
- 8 In $\triangle ABC$ below, $\overleftrightarrow{CE}$ is the perpendicular bisector of $\overline{AB}$.

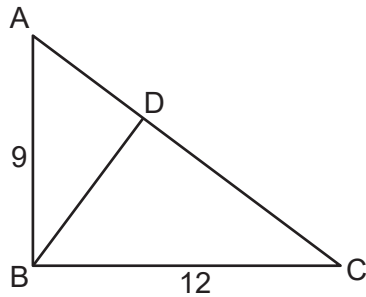


Which statement is always true?

- (1) $\overline{AC} \cong \overline{BC}$ (3) $\angle EAC \cong \angle BCE$
(2) $\overline{AE} \cong \overline{CE}$ (4) $m\angle A + m\angle B = 90^\circ$

Use this space for
computations.

- 9 In right triangle ABC below, $AB = 9$, $BC = 12$, and altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$.



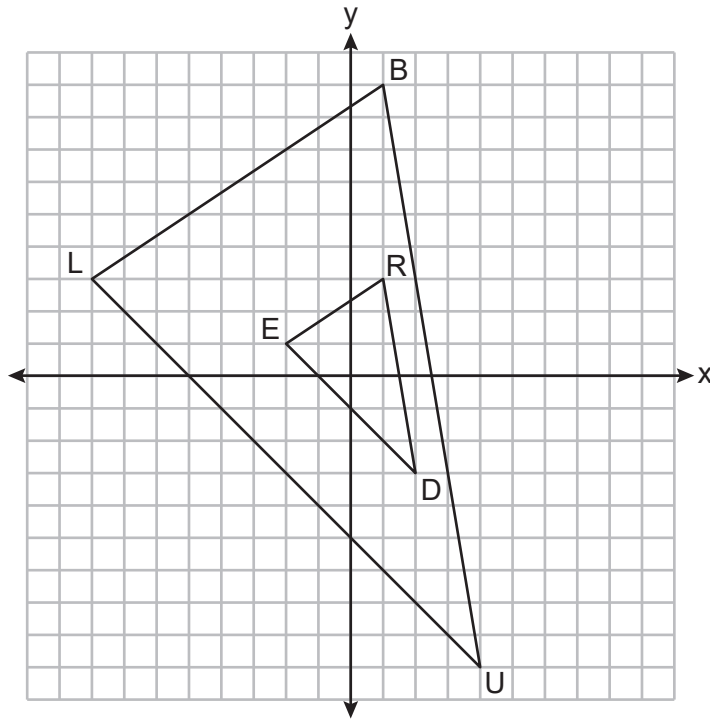
Which equation is always true for $\overline{BD}$?

- (1) $\cos A = \frac{BD}{9}$ (3) $\tan A = \frac{BD}{9}$
(2) $\sin C = \frac{BD}{12}$ (4) $\sin C = \frac{BD}{15}$
- 10 Which regular polygon, when rotated about its center, carries onto itself after both a 120° rotation and a 180° rotation?
- (1) triangle (3) hexagon
(2) square (4) octagon
- 11 The coordinates of the endpoints of $\overline{PA}$ are $P(3, -6)$ and $A(-2, 9)$. If point C is on $\overline{PA}$, what are the coordinates of C such that $PC:CA = 1:4$?

- (1) $(-1, 6)$ (3) $(1, 0)$
(2) $(0, 3)$ (4) $(2, -3)$

Use this space for
computations.

12 On the set of axes below, $\triangle BLU$ is the image of $\triangle RED$ after a dilation.



What are the scale factor and the coordinates of the center of dilation of this transformation?

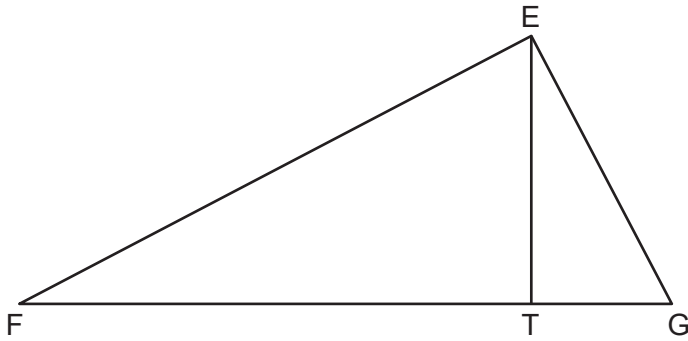
- | | |
|------------------|------------------|
| (1) 2 and (0, 0) | (3) 3 and (0, 0) |
| (2) 2 and (1, 0) | (4) 3 and (1, 0) |

13 What are the coordinates of the center and the length of the radius of the circle whose equation is $x^2 + y^2 = 45 + 4x$?

- (1) center (2, 0) and radius 7
- (2) center (−2, 0) and radius 7
- (3) center (2, 0) and radius 49
- (4) center (−2, 0) and radius 49

Use this space for
computations.

- 14 In right triangle EFG below, altitude $\overline{ET}$ is drawn to hypotenuse $\overline{FG}$.



If $EF = 17$ and $FT = 15$, what is the length of $\overline{TG}$, to the nearest tenth?

- (1) 3.8 (3) 8.0
(2) 4.3 (4) 9.1
- 15 In trapezoid $ERJT$, sides $\overline{ER}$ and $\overline{TJ}$ are parallel.
If $m\angle R = (2x + 15)^\circ$ and $m\angle J = (3x - 40)^\circ$, what is $m\angle J$?

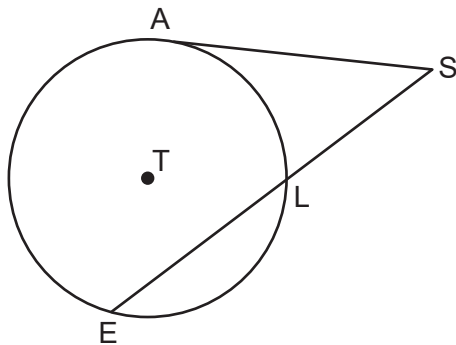
- (1) 125° (3) 83°
(2) 97° (4) 55°

Use this space for
computations.

- 16 In isosceles right triangle ABC , the length of hypotenuse $\overline{AC}$ is 14. What is the length of $\overline{BC}$, to the *nearest tenth*?

- (1) 7.0 (3) 9.9
(2) 8.1 (4) 19.8

- 17 In circle T below, tangent $\overline{AS}$ and secant $\overline{ELS}$ are drawn.



If $SL = 8$ and $LE = 10$, the length of $\overline{AS}$ is

- (1) $\sqrt{18}$ (3) 9
(2) $\sqrt{80}$ (4) 12

- 18 Parallelogram $MERT$ has diagonals that intersect at I . Which additional statement is sufficient to prove $MERT$ is a rhombus?

- (1) $\angle ERT \cong \angle RTM$ (3) $m\angle TIM = 90^\circ$
(2) $\angle MEI \cong \angle RTI$ (4) $\overline{ME} \cong \overline{RT}$

Use this space for
computations.

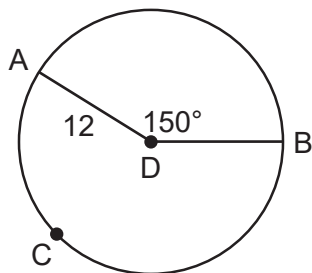
- 19 For the acute angles in right triangle ABC , $\sin(3x)^\circ = \cos(x + 10)^\circ$.
What is the measure of the *smallest* angle in $\triangle ABC$?

- (1) 5° (3) 30°
(2) 15° (4) 60°

- 20 Which two-dimensional shape below can *not* be a plane section of a rectangular prism?

- (1) triangle (3) pentagon
(2) octagon (4) trapezoid

- 21 Points A , B , and C are on circle D below such that $DA = 12$ and $m\angle ADB = 150^\circ$.



The length of $\widehat{AB}$ is

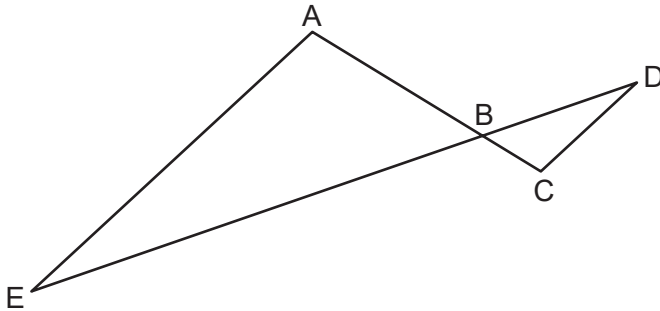
- (1) 5π (3) 24π
(2) 10π (4) 60π

Use this space for
computations.

- 22** The lengths of two sides of a triangle are 12 and 30.
The length of the third side could be

- (1) 12 (3) 28
(2) 18 (4) 42

- 23** In the diagram below, $\overline{AC}$ and $\overline{ED}$ intersect at B , and $\overline{AE} \parallel \overline{CD}$.



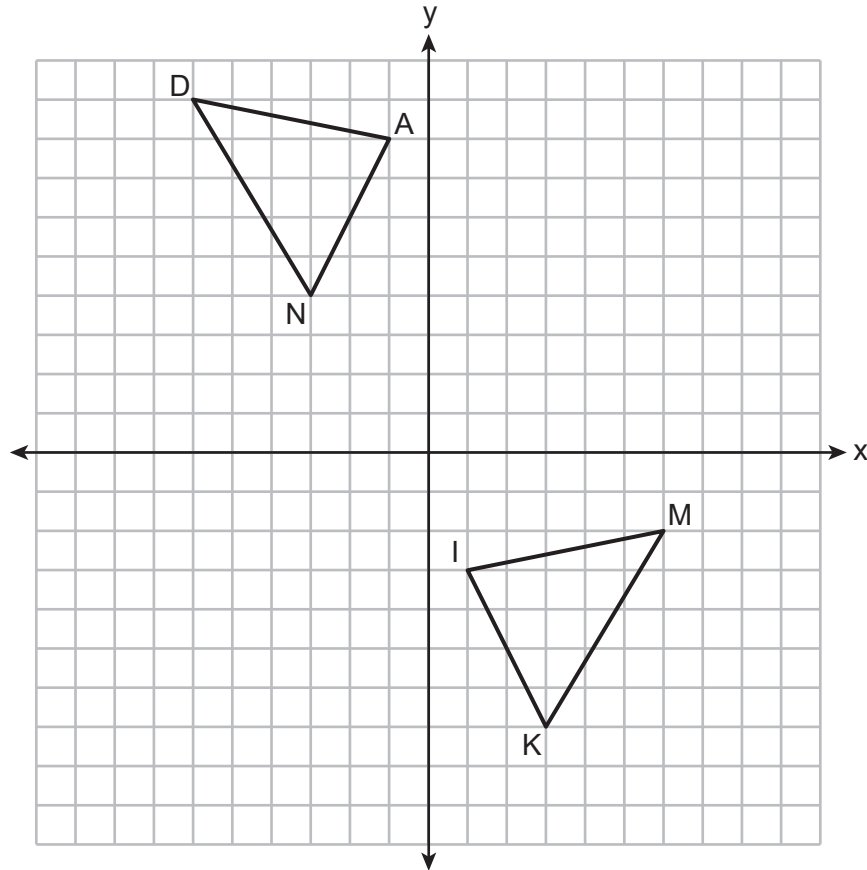
If $AB = 6$, $BC = 2$, $CD = 3$, and $BD = 4$, what is the perimeter of $\triangle ABE$?

- (1) 9 (3) 21
(2) 18 (4) 27
- 24** On the coordinate plane, a line is dilated by a scale factor, centered at a point on the line. The image of the line has
- (1) the same slope and the same y -intercept as the original line
 - (2) the same y -intercept but a different slope as the original line
 - (3) the same slope but a different y -intercept as the original line
 - (4) a different slope and a different y -intercept than the original line
-

Part II

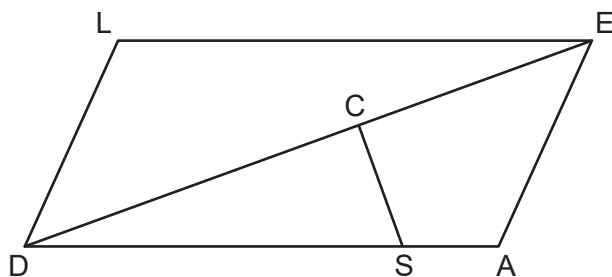
Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

25 On the set of axes below, $\triangle DAN \cong \triangle MIK$.



Describe a sequence of rigid motions that maps $\triangle DAN$ onto $\triangle MIK$.

26 In parallelogram $LEAD$ below, C is on $\overline{DE}$ and S is on $\overline{DA}$ such that $\overline{SC} \perp \overline{DE}$.

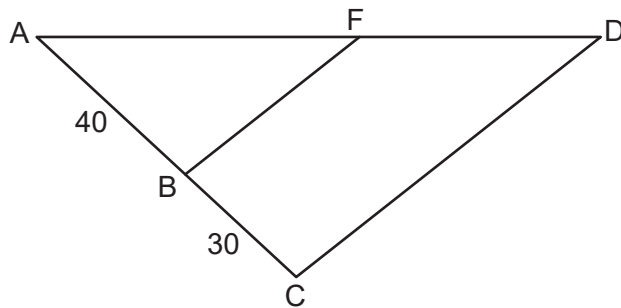


If $m\angle LED = 21^\circ$, determine and state the measure of $\angle CSA$.

27 A section of tree trunk from a white pine tree can be modeled by a cylinder. The diameter of the trunk is 1.5 feet, and the height of this section is 8 feet.

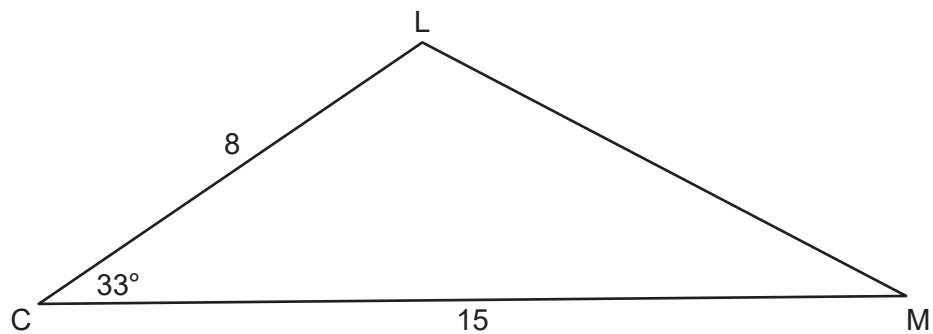
If white pine weighs 25 pounds per cubic foot, determine and state the weight of this section of the white pine tree, to the *nearest pound*.

- 28** In $\triangle ADC$ below, points B and F are on $\overline{AC}$ and $\overline{AD}$, respectively, such that $AB = 40$, $BC = 30$, and $\overline{BF}$ is drawn.



If $AF = 64$, determine and state the length of $\overline{FD}$ that would prove $\overline{BF} \parallel \overline{CD}$.

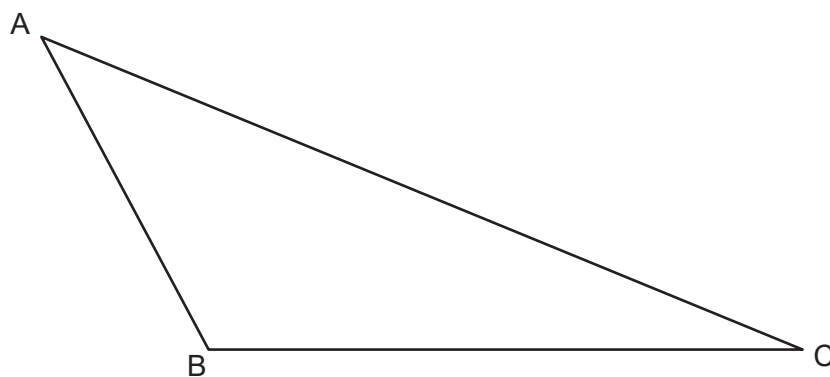
29 In $\triangle CLM$ below, $m\angle C = 33^\circ$, $CL = 8$, and $CM = 15$.



Determine and state, to the *nearest tenth*, the area of $\triangle CLM$.

30 In the diagram of $\triangle ABC$ below, use a compass and straightedge to construct the angle bisector of $\angle ABC$.

[Leave all construction marks.]



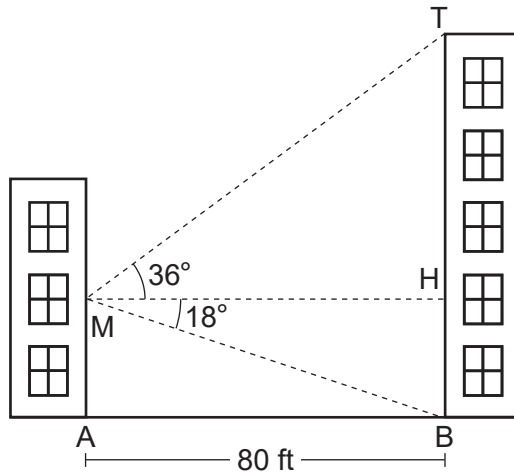
31 The diagonals of parallelogram $GRAM$ intersect at P .

If $RP = 12$ and $GA = 24$, explain why $GRAM$ is a rectangle.

Part III

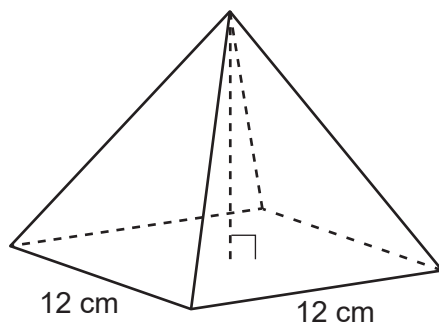
Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

- 32 As modeled below, Maria wants to determine the height of the building across the street from her position, M . The angle of elevation from M to the top of the building, T , is 36° . From M , the angle of depression to the base of the building, B , is 18° . The buildings are 80 feet apart.



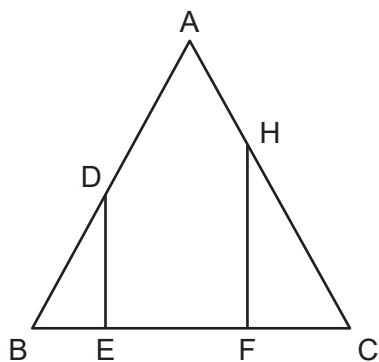
Determine and state, to the *nearest foot*, the height of the building, TB , across the street from Maria.

- 33** An artist uses clay to make solid pyramids with a square base whose sides measure 12 cm, as modeled below.



The height of each pyramid is 8 cm, and the density of the clay is 1.25 g/cm^3 . If the artist has 15 kilograms of clay, determine and state the maximum number of pyramids that can be made.

34 Given: Triangle ABC , $\overline{AB} \cong \overline{AC}$, $\overline{DE} \perp \overline{BC}$, and $\overline{HF} \perp \overline{BC}$



Prove: $\frac{DE}{BE} = \frac{HF}{CF}$

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

35 Triangle ABC has vertices whose coordinates are $A(-3, -1)$, $B(-5, 2)$, and $C(-1, 8)$.

State the coordinates of point D such that quadrilateral $ABCD$ is a parallelogram.

[The use of the set of axes on the next page is optional.]

Prove $ABCD$ is a parallelogram.

[The use of the set of axes on the next page is optional.]

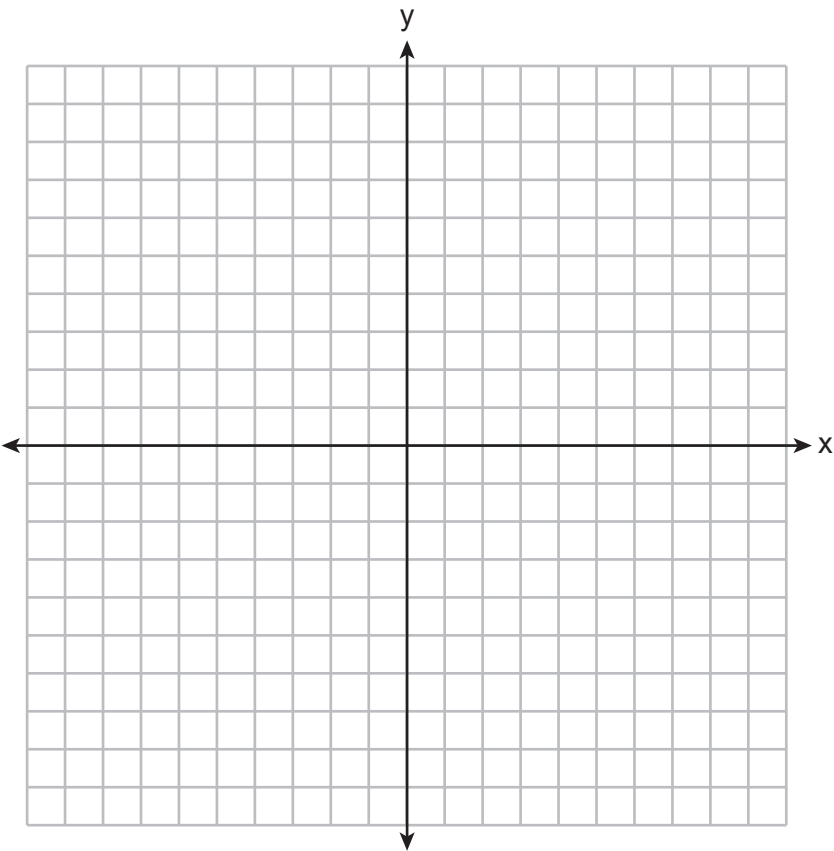
State the coordinates of point E , the midpoint of $\overline{BC}$.

[The use of the set of axes on the next page is optional.]

Question 35 is continued on the next page.

Question 35 continued

Prove $\triangle ABE$ is an isosceles triangle.
[The use of the set of axes below is optional.]



Regents Examination in Geometry – June 2026**Scoring Key: Part I (Multiple-Choice Questions)**

Examination	Date	Question Number	Scoring Key	Question Type	Credit
Geometry	June '26	1	4	MC	2
Geometry	June '26	2	1	MC	2
Geometry	June '26	3	3	MC	2
Geometry	June '26	4	2	MC	2
Geometry	June '26	5	4	MC	2
Geometry	June '26	6	1	MC	2
Geometry	June '26	7	2	MC	2
Geometry	June '26	8	1	MC	2
Geometry	June '26	9	2	MC	2
Geometry	June '26	10	3	MC	2
Geometry	June '26	11	4	MC	2
Geometry	June '26	12	4	MC	2
Geometry	June '26	13	1	MC	2
Geometry	June '26	14	2	MC	2
Geometry	June '26	15	3	MC	2
Geometry	June '26	16	3	MC	2
Geometry	June '26	17	4	MC	2
Geometry	June '26	18	3	MC	2
Geometry	June '26	19	3	MC	2
Geometry	June '26	20	2	MC	2
Geometry	June '26	21	2	MC	2
Geometry	June '26	22	3	MC	2
Geometry	June '26	23	4	MC	2
Geometry	June '26	24	1	MC	2

Regents Examination in Geometry – June 2026**Scoring Key: Parts II, III, and IV (Constructed-Response Questions)**

Examination	Date	Question Number	Scoring Key	Question Type	Credit
Geometry	June '26	25	-	CR	2
Geometry	June '26	26	-	CR	2
Geometry	June '26	27	-	CR	2
Geometry	June '26	28	-	CR	2
Geometry	June '26	29	-	CR	2
Geometry	June '26	30	-	CR	2
Geometry	June '26	31	-	CR	2
Geometry	June '26	32	-	CR	4
Geometry	June '26	33	-	CR	4
Geometry	June '26	34	-	CR	4
Geometry	June '26	35	-	CR	6

Key

MC = Multiple-choice question
 CR = Constructed-response question

The chart for determining students' final examination scores for the **June 2026 Regents Examination in Geometry** will be posted on the Department's website at: <https://www.nysedregents.org/geometryre/> on the day of the examination. Conversion charts provided for the previous administrations of the Regents Examination in Geometry must NOT be used to determine students' final scores for this administration.